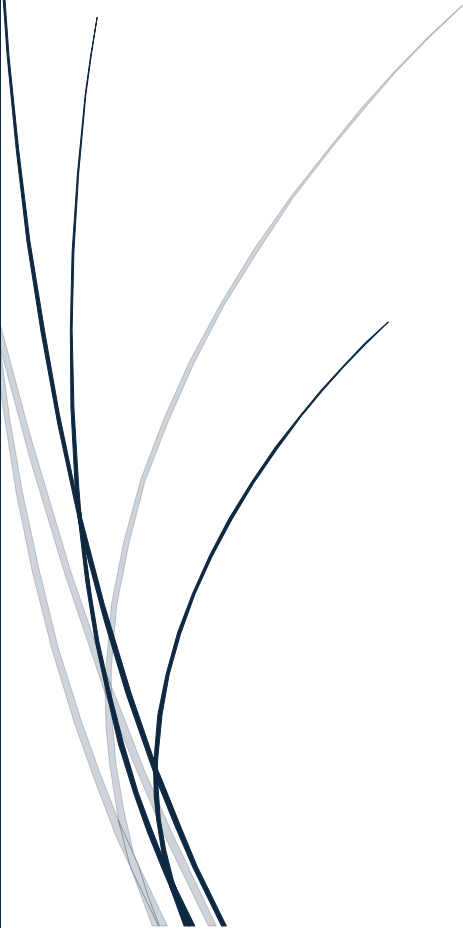




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Interactive Plotly/Dash dashboard
Dataset:- IAPT and Open Prescribing Dataset (NHS England)



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DATA ANALYTICS INTERN

Introduction

This project developed an interactive Plotly Dash dashboard to visualise NHS mental health analytics using:

- IAPT (Improving Access to Psychological Therapies) data
- NHS Open Prescribing data

The dashboard integrates treatment pathway metrics, prescribing trends, regional comparisons, and forecasting into a single interactive analytics platform.

The application was designed as a localhost demonstration dashboard using Python Dash and Plotly.

Project Objectives

The primary objectives of the dashboard were:

1. Build an interactive healthcare analytics dashboard.
2. Visualise prescribing trends for antidepressant and anxiolytic medicines.
3. Monitor IAPT operational performance indicators.
4. Analyse treatment drop-off and patient funnel progression.
5. Support region-level and category-level filtering using dropdowns.
6. Provide dashboard-ready NHS mental health analytics.

Technologies Used

Technology	Purpose
Python	Data processing
Pandas	Data manipulation
Plotly	Interactive visualisations
Dash	Dashboard framework
OpenPyXL	Excel export
Matplotlib	Exploratory charts

Datasets Used

Two primary NHS datasets were integrated into the dashboard.

NHS Open Prescribing Dataset

Used for:

- antidepressant prescribing analysis,
- anxiolytic prescribing analysis,
- prescribing cost trends,
- medicine-level analysis.

Important Variables

Variable	Description
YEAR_MONTH	Prescribing month
ICB_NAME	Integrated Care Board
BNF_CHEMICAL_SUBSTANCE	Medicine name
ITEMS	Prescribed items
TOTAL_QUANTITY	Prescribing quantity
ACTUAL_COST	Prescribing cost

NHS Talking Therapies / IAPT Dataset

Used for:

- RTT waiting-time analysis,
- treatment funnel analysis,
- drop-off analysis,
- estimated DNA analysis.

Important Variables

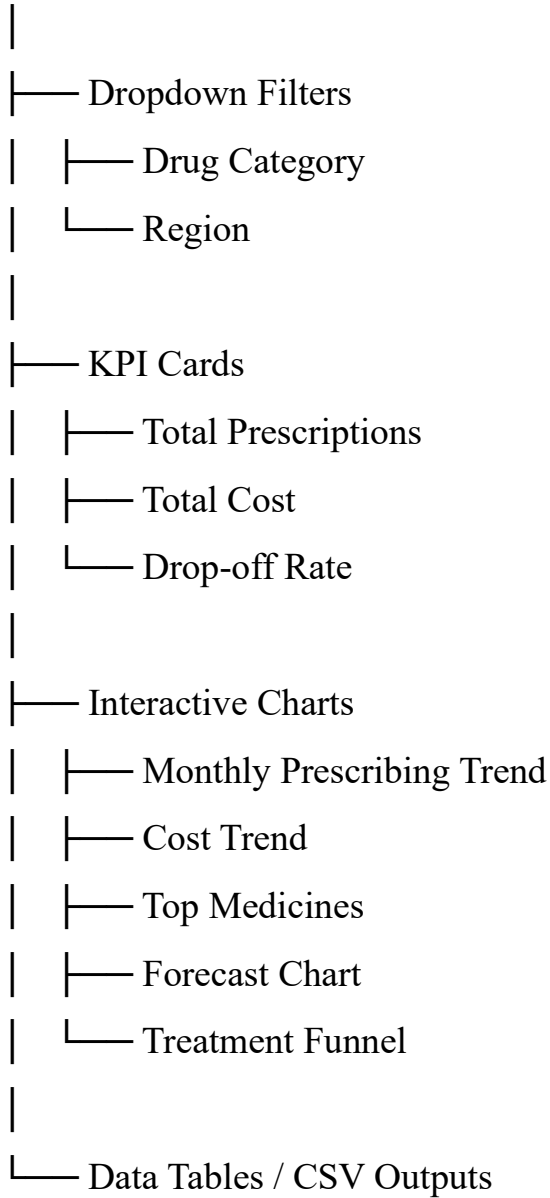
Variable	Description
Count_AccessingServices	Service access count
Count_FinishedCourseTreatment	Completed treatment
Mean_WaitAccessingServices	RTT waiting time

Dashboard Architecture

The dashboard was designed using a modular structure.

Main Components

Dashboard Layout



Dashboard Features

Interactive Dropdown Filters

The dashboard included dynamic filtering functionality:

Filter	Purpose
Drug Category Dropdown	Switch between antidepressants and anxiolytics
Region Dropdown	Filter by ICB region

All charts automatically updated when filters changed.

KPI Cards

The dashboard displayed real-time KPI metrics:

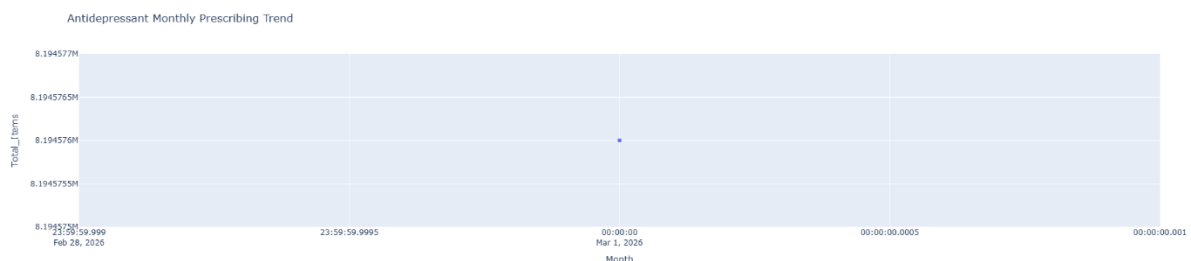
KPI	Description
Total Prescriptions	Sum of prescribed items
Total Cost	Total NHS prescribing expenditure
Drop-off Rate	Estimated treatment disengagement

These KPI cards provided a quick operational summary.

Visualisations Implemented

Monthly Prescribing Trend

Interactive line charts displayed monthly prescribing behaviour across mental health medicine categories.

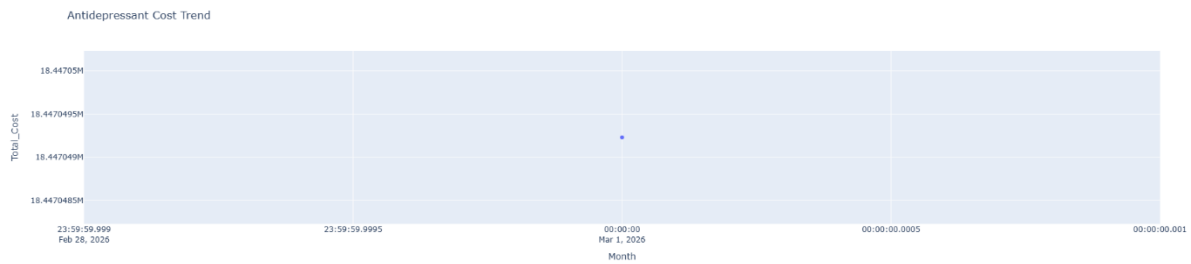


Insights

- Antidepressant prescribing was consistently higher than anxiolytics.
- Prescribing demand remained relatively stable over time.

Prescribing Cost Trend

Cost trend charts visualised prescribing expenditure over time.

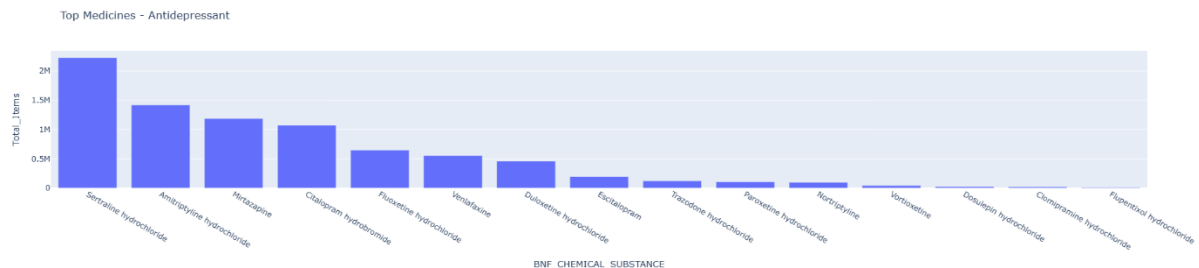


Insights

- NHS mental health prescribing generated significant expenditure.
- Cost fluctuations followed prescribing volume changes.

Top Medicines Analysis

Bar charts highlighted the most prescribed mental health medicines.



Frequently Prescribed Medicines

- Sertraline
- Fluoxetine
- Citalopram
- Diazepam
- Mirtazapine

Forecasting Visualisation

A 3-month moving-average forecasting model estimated short-term prescribing demand.

Purpose

- Support operational planning
- Estimate future prescribing volume
- Visualise expected demand continuation

Treatment Funnel Analysis

A funnel chart visualised patient progression through treatment stages.



Funnel Stages

Accessing Services



Finished Treatment



Data Processing Workflow

The dashboard backend followed these steps:

1. Load CSV summary datasets
2. Clean and standardise data
3. Apply dropdown filters
4. Generate Plotly figures dynamically
5. Update dashboard outputs using Dash callbacks

Forecasting Methodology

Forecasting used a simple moving-average method.

The model:

- calculated rolling averages,
- projected future prescribing items,
- and estimated short-term demand patterns.

Although simple, the method provided interpretable operational forecasts.

Dashboard Outputs

The project generated several dashboard-ready datasets.

Output File	Purpose
dashboard_monthly_trend.csv	Monthly prescribing trends
dashboard_icb_summary.csv	ICB-level summary
dashboard_drug_summary.csv	Medicine analysis
prescribing_3_month_forecast.csv	Forecasting
treatment_funnel_summary.csv	Funnel analysis
dropoff_rate_by_region.csv	Drop-off analysis

Localhost Deployment

The dashboard was deployed locally using:

python app.py

and accessed through:

http://127.0.0.1:8050

The dashboard successfully supported real-time interaction and chart updates.

Key Findings

The dashboard analysis identified several important insights:

- Antidepressants represented the dominant prescribing category.
- Mental health prescribing costs were substantial across NHS regions.
- Treatment drop-off remained a significant operational issue.

- Prescribing demand remained stable with gradual upward trends.
- Regional variation existed across ICBs.
- Funnel analysis demonstrated measurable patient attrition.

Limitations

This dashboard has several limitations:

1. Forecasting used a basic moving-average model.
2. Population-adjusted prescribing rates were not included.
3. Real-time NHS API integration was not implemented.
4. Some DNA metrics were estimated due to unavailable direct variables.

Conclusion

This project successfully developed an interactive NHS mental health analytics dashboard integrating:

- prescribing trends,
- IAPT operational metrics,
- forecasting,
- and treatment pathway analytics.

The dashboard demonstrates how Python Dash and Plotly can support healthcare analytics, operational monitoring, and mental health service planning.

The system provides a scalable foundation for future enhancements such as:

- real-time NHS integration,
- advanced forecasting,
- geographic mapping,
- and machine learning-based prediction systems.